A compact subshift without a harmonic entropy convergence bound

Submission for TLMC problem 00000008488

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Claim and scope. The conjecture includes a uniform $1/n$ convergence rate for finite-scale values in the equivalent definitions of Bowen topological entropy. We use the standard convention that n is the orbit length and a finite-scale value is $n^{-1}\log r_n(U)$, where $r_n(U)$ is the minimum size of a spanning cover at a fixed entourage U. We give a compact continuous dynamical system for which even one fixed entourage admits no eventual C/n error bound, for any constant C. This refutes that rate assertion under this explicit conventional interpretation. We do not dispute the classical equality of entropy definitions on compact systems, and make no claim about the unspecified noncompact boundary correction or periodic optimality clauses.

The actual compact system. Give $\{0,1\}^{\mathbb{N}}$ its product topology and usual compatible metric

$$d(x,y)=\begin{cases}0, & x=y,\\ 2^{-\min\{j:x_j\neq y_j\}}, & x\neq y.\end{cases}$$

Let

$$X=\{x\in\{0,1\}^{\mathbb{N}}:\text{at most one coordinate of } x \text{ is } 1\},\qquad (Tx)_j=x_{j+1}.$$

The condition defining X is the intersection, over distinct i,j, of the closed conditions $x_i=0$ or $x_j=0$. Hence X is a closed subset of a compact product and is compact. The left shift preserves X and is continuous, as each output coordinate is an input coordinate. Write z for the all-zero sequence and e_j for the sequence whose unique 1 occurs at j.

For $m\geq 0$ let U_m mean agreement on the first m coordinates. These are entourages, and every entourage contains some U_m. Indeed, agreement on the first m coordinates is equivalent to $d(x,y)\leq 2^{-m}$, and 2^{-m} tends to zero. Thus this family is cofinal in the actual metric uniformity of X.

Exact spanning counts at a fixed scale. For an entourage U set

$$U^{(n)}=\{(x,y):(T^kx,T^ky)\in U \text{ for } 0\leq k<n\}.$$

The number $r_n(U)$ is the minimum cardinality of a finite set of centres whose $U^{(n)}$-balls cover X. These are precisely the objects `Dynamics.dynEntourage` and `Dynamics.coverMincard` used in Lean. The actual iterates satisfy $(T^kx)_i=x_{i+k}$, so

$$U_1^{(n)}=U_n.$$

The $n+1$ points

$$S_n=\{z,e_0,\dots,e_{n-1}\}$$

give a spanning cover: a point with its 1 before coordinate n is represented by that e_j, and every other point is represented by z. Their U_n-balls are pairwise disjoint. Consequently every spanning cover needs at least $n+1$ centres, and

$$r_n(U_1)=n+1.$$

The formal proof constructs S_n from the injective map $\mathrm{Option}(\mathrm{Fin}\, n) \to X$ and proves its cardinality; the covering number is not assigned as a definition. It also proves this covering number is finite before its natural-number value is used.

The full topological entropy is zero. For any fixed m, the same construction with S_{m+n} spans at scale U_m for n iterates. Thus

$$1 \le r_n(U_m) \le m + n + 1, \qquad \lim_{n\to\infty} \frac{\log(m+n+1)}{n} = 0.$$

It follows that the entropy at every U_m is zero. Since the U_m are cofinal, the supremum over all entourages is also zero:

$$h_{\mathrm{top}}(T) = 0.$$

Lean proves this equality for the native Bowen–Dinaburg quantity `Dynamics.coverEntropy`; it is not merely a limit of an independently chosen numerical sequence. The standard symbolic entropy convention is also recalled in [1, p. 1], but no external entropy evaluation is assumed in the formal proof.

Failure of every harmonic bound. For $n \ge 1$, the actual finite-time value at U_1 is

$$a_n = \frac{\log r_n(U_1)}{n} = \frac{\log(n+1)}{n}.$$

Although $a_n \to h_{\mathrm{top}}(T) = 0$, we have $na_n = \log(n+1) \to \infty$. More explicitly, for every $C \in \mathbb{R}$ and $N \in \mathbb{N}$ choose $n \ge \max\{N, 1\}$ with $\log(n+1) > C$. Then

$$|a_n - h_{\mathrm{top}}(T)| = \frac{\log(n+1)}{n} > \frac{C}{n}.$$

This is the full quantified obstruction to an eventual $O(1/n)$ bound, even at this single fixed scale. A bound uniform over scales or across the stated definitions therefore cannot hold.

Formal files. The complete construction and proof are in `lean/Main.lean`; the conclusion is `TLMC8488.counterexample`. The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

The full original bilingual source is preserved in `SOURCE.md`. Build commands, exhaustive namespace axiom checks, fresh-kernel receipts and file hashes accompany the submission.

References

[1] A. Mitchell, *On word complexity and topological entropy of random substitution subshifts*, arXiv:2305.04817v2 (2024). arxiv.org/abs/2305.04817.